



GIRRAWEE HIGH SCHOOL

YEAR 11 - TASK 4

2007

MATHEMATICS

2 UNIT

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new sheet of paper.

Question 1 (11 marks)

Marks

- (a) Find the distance between the points A(3, 1) and B(2, -1), leave answer in surd form. 2
- (b) Find the equation of the line joining the points (4, 8) and (2, 16). 3
- (c) If (2, 3) is the midpoint of the two points A(x, y) and B(4, 4). Find the coordinates of A. 3
- (d) A line is represented by the equation $\frac{1}{2}x + y - 3 = 0$. Find the equation of the line which is perpendicular to this line and that passes through the origin. 3

Question 2 (11 marks)

- (a) Find the perpendicular distance from the point (4, 5) to the line $x + 2y - 4 = 0$. 2
- (b) Find the angle of the line $y = -x + 2$ makes with the x-axis. 2
- (c) Find the equation of the line that passes through the point of intersection of the two lines $y = 2x + 1$ and $y = -2x + 5$, that has a gradient of 4. 4
- (d) Find the equation of the line that passes through the points (0,3) and (s, t), in gradient-intercept form. 3

Question 3 (21 marks)

(a) Differentiate

(i) $y = \frac{1}{x^3} + \sqrt{x}$ 3

(ii) $y = (3x^2 + 1)(5x^3 + 2)$ 3

(iii) $y = \frac{2x + 5}{x + 2}$ 3

(iv) $y = \frac{x^2}{4 - x}$ 3

(b) Evaluate

(i) $\lim_{x \rightarrow 0} \left(\frac{4x^2}{x^4 + x^2} \right)$ 3

(ii) $\lim_{x \rightarrow 3} \left(\frac{x^2 - 9}{x - 3} \right)$ 3

(iii) $\lim_{x \rightarrow \infty} \left(\frac{3x^2 - 15x - 19}{5x^2 + 13x + 1} \right)$ 3

Question 4 (20 marks)

(a) Find $\frac{dy}{dx}$.

(i) $y = \sqrt[3]{x^2} - \frac{1}{x}$ 3

(ii) $y = \sqrt{3x + 5}$ 3

(iii) $y = \frac{2}{(1 - 7x)^3}$ 3

(iv) $y = (5x^3 + 4x)^{\frac{3}{2}}$ 3

(v) $y = x^3(6x^2 + 1)^5$ 4

(vi) $y = \frac{2x}{\sqrt[3]{x^2 + 1}}$ 4

Question 5 (11 marks)

- (a) Find the gradient of the curve $y = 15x^4 + 7x^2$ at the point where $x = 9$.

3

- (b) Find the equation of the tangent to the curve $y = x^2$ at the point (3, 9).

3

- (c) Find the points where $f'(x) = 0$, for the function

$$f(x) = \frac{1}{3}x^3 - 5x^2 + 16x + 5$$

5

Question 6 (11 marks)

- (a) Given $y = \frac{1}{\sqrt{x^2 + 9}}$, find the equation of the normal of the curve at $x = 4$.

7

- (b) Using first principles, find the derivative of $y = x^2 - 4x + 4$.

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* Mathematics (2007)
Year 11 - Task 4

Q1

$$a) d_{AB} = \sqrt{(3-2)^2 + (1-(-1))^2}$$

$$= \sqrt{1+4}$$

$$d = \sqrt{5}$$

$$b) m = \frac{16-8}{2-4} = \frac{8}{-2} = -4$$

$$(4, 8), m = -4$$

$$y - 8 = -4(x - 4)$$

$$y - 8 = -4x + 16$$

$$y = -4x + 24$$

$$\text{or } 4x + y - 24 = 0$$

$$c) 2 = \frac{x+4}{2} ; 3 = \frac{y+4}{2}$$

$$4 = x + 4$$

$$x = 0$$

$$6 = y + 4$$

$$y = 2$$

$$d) \frac{1}{2}x + y - 3 = 0$$

$$y = -\frac{1}{2}x + 3$$

$$\therefore m_1 = -\frac{1}{2} \text{ and } m_2 = +2$$

$$\therefore m = 2, (0, 0)$$

$$y = 2x$$

$$Q2) a) d_{\perp} = \frac{|4 + 2(5) - 4|}{\sqrt{1^2 + 2^2}}$$

$$= \frac{10}{\sqrt{5}} = 2\sqrt{5}$$

$$b) m = -1$$

$$\tan \theta = m$$

$$\tan \theta = -1$$

$$\theta = -45^\circ \text{ or } 135^\circ$$

$$Q2) a) y = 2x + 1 \text{ ①; } y = -2x + 5 \text{ ②}$$

$\therefore$ Sub ① into ②

$$2x + 1 = -2x + 5$$

$$4x = 4$$

$$x = 1$$

$$\therefore y = 2(1) + 1$$

$$y = 3 \quad (1, 3) \text{ intersection point.}$$

$$m = 4$$

$$\therefore y - 3 = 4(x - 1)$$

$$y - 3 = 4x - 4$$

$$y = 4x - 1$$

$$\text{or } 4x - y - 1 = 0$$

Q3

$$(a) (i) y = x^{-3} + x^{\frac{1}{2}}$$

$$y' = -3x^{-4} + \frac{1}{2}x^{-\frac{1}{2}}$$

$$y' = \frac{-3}{x^4} + \frac{1}{2\sqrt{x}}$$

$$(ii) y = 15x^5 + 5x^3 + 6x^2 + 2$$

$$y' = 75x^4 + 15x^2 + 12x$$

$$y' = 3x(25x^3 + 5x + 4)$$

$$(iii) y = \frac{2x+5}{x+2}$$

$$y' = \frac{(x+2)(2) - (2x+5)(1)}{(x+2)^2}$$

$$= \frac{2x+4-2x-5}{(x+2)^2}$$

$$y' = \frac{-1}{(x+2)^2}$$

$$(iv) y = \frac{x^2}{4-x}$$

$$y' = \frac{(4-x)(2x) - x^2(-1)}{(4-x)^2}$$

$$= \frac{8x - 2x^2 + x^2}{(4-x)^2}$$

$$y' = \frac{x(8-x)}{(4-x)^2}$$

$$(b) \lim_{x \rightarrow 0} \frac{4x^2}{x^2(x^2+1)}$$

$$= \lim_{x \rightarrow 0} \frac{4}{x^2+1}$$

$$= \frac{4}{0^2+1}$$

$$= 4$$

$$(ii) \lim_{x \rightarrow 3} \frac{(x+3)(x-3)}{(x-3)}$$

$$= \lim_{x \rightarrow 3} (x+3)$$

$$= 3+3$$

$$= 6$$

$$(iii) \lim_{x \rightarrow \infty} \frac{3 - \frac{15}{x} - \frac{19}{x^2}}{5 + \frac{13}{x} + \frac{1}{x^2}}$$

$$= \frac{3 - 0 - 0}{5 + 0 + 0}$$

$$= \frac{3}{5}$$

Q4

$$(a) (i) y = x^{\frac{2}{3}} - x^{-1}$$

$$\frac{dy}{dx} = \frac{2}{3}x^{-\frac{1}{3}} + x^{-2}$$

$$\frac{dy}{dx} = \frac{2}{3\sqrt[3]{x}} + \frac{1}{x^2}$$

$$(ii) y = (3x+5)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(3x+5)^{-\frac{1}{2}} \times 3$$

$$\frac{dy}{dx} = \frac{3}{2\sqrt{3x+5}}$$

$$(iii) y = 2(1-7x)^3$$

$$\frac{dy}{dx} = -6(1-7x)^2 \times (-7)$$

$$\frac{dy}{dx} = \frac{42}{(1-7x)^2}$$

$$(iv) y = (5x^3+4x)^{\frac{3}{2}}$$

$$\frac{dy}{dx} = \frac{3}{2}(5x^3+4x)^{\frac{1}{2}} \times (15x^2+4)$$

$$\frac{dy}{dx} = \frac{3}{2}(15x^2+4)(5x^3+4x)^{\frac{1}{2}}$$

$$(v) y = x^3(6x^2+1)^5$$

$$\frac{dy}{dx} = (6x^2+1)^5(3x^2) + (x^3)5(6x^2+1)^4 \times 12$$

$$= 3x^2(6x^2+1)^5 + 60x^4(6x^2+1)^4$$

$$= 3x^2(6x^2+1)^4[6x^2+1 + 20x^2]$$

$$= 3x^2(26x^2+1)(6x^2+1)^4$$

$$(vi) y = \frac{2x}{(x^2+1)^{\frac{1}{3}}}$$

$$\frac{dy}{dx} = \frac{(x^2+1)^{\frac{1}{3}}(2) - (2x)\frac{1}{3}(x^2+1)^{-\frac{2}{3}} \times 2x}{((x^2+1)^{\frac{1}{3}})^2}$$

$$\frac{dy}{dx} = \frac{2(x^2+1)^{\frac{1}{3}} - \frac{4}{3}x^2(x^2+1)^{-\frac{2}{3}}}{(x^2+1)^{\frac{2}{3}}}$$

Q5 (a) $y = 15x^4 + 7x^2$
 $y' = 60x^3 + 14x$
 when $x=9$ $y' = 60(9)^3 + 14(9)$
 $y' = 43740 + 126$
 $y' = 43866$

(b) $y = x^2$
 $\frac{dy}{dx} = 2x$

when $x=3$ $\frac{dy}{dx} = 2(3)$
 $(2-9) \quad m = 6$
 $y - 9 = 6(x - 3)$
 $y - 9 = 6x - 18$
 $y = 6x - 9$

(c) $f(x) = \frac{1}{3}x^3 - 5x^2 + 16x + 5$

$f'(x) = x^2 - 10x + 16$

$f'(x) = (x-8)(x-2)$

when $f'(x) = 0$ $0 = (x-8)(x-2)$
 $\therefore x = 8, 2$

$f(8) = \frac{1}{3}(8)^3 - 5(8)^2 + 16(8) + 5$
 $= -16\frac{1}{3} \quad \therefore (8, -16\frac{1}{3})$

$f(2) = \frac{1}{3}(2)^3 - 5(2)^2 + 16(2) + 5$
 $= 19\frac{2}{3} \quad \& (2, 19\frac{2}{3})$

Q6 $y = (x^2 + 9)^{-\frac{1}{2}}$
 $y' = -\frac{1}{2}(x^2 + 9)^{-\frac{3}{2}} \times 2x$
 $y' = \frac{-x}{(x^2 + 9)^{\frac{3}{2}}}$

when $x=4$ $y' = \frac{-4}{(4^2 + 9)^{\frac{3}{2}}}$ & $y = (4^2 + 9)^{-\frac{1}{2}}$
 $y = \frac{1}{5}$

$y' = -\frac{4}{125}$ perpendicular line

$\therefore m_1 = -\frac{4}{125} \quad \therefore m_2 = \frac{125}{4}$

$(4, \frac{1}{5})$

$y - \frac{1}{5} = \frac{125}{4}(x - 4)$

$y - \frac{1}{5} = \frac{125x}{4} - 125$

$y = \frac{125}{4}x - \frac{624}{5}$

20y = 625x - 2496

625x - 20y - 2496 = 0

b) $f(x) = x^2 - 4x + 4$

$f(x+h) = (x+h)^2 - 4(x+h) + 4$

$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 4(x+h) + 4 - (x^2 - 4x + 4)}{h}$

$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4x - 4h + 4 - x^2 + 4x - 4}{h}$

$= \lim_{h \rightarrow 0} \frac{h(2x - 4 + h)}{h}$

$= 2x - 4 + 0$

$\frac{dy}{dx} = 2x - 4$